

Competitive Exam Practice 9 — Answer Key

Cache Memory

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- Q1 (D).** Locality — temporal (a location referenced now is likely referenced again soon) and spatial (nearby locations are likely referenced soon) — is the property that lets a small, fast cache stand in for a large, slow main memory. Interrupts, DMA and polling are I/O techniques; none is justified by locality.
- Q2 (D).** Fully associative mapping has *no index field* (0 index bits): a block may go in any line, identified purely by tag comparison. The block size $16\text{ B} = 2^4$ fixes the offset at 4 bits, so tag = $32 - 4 = 28$ bits. (The cache holds $16\text{ kB}/16\text{ B} = 1024$ lines, but a line is selected by comparison, never by an index.)
- Q3 (A).** Block = $32\text{ B} = 2^5 \Rightarrow$ word field 5 bits. Lines = $32\text{ KB}/32\text{ B} = 1024 = 2^{10} \Rightarrow$ index 10 bits. Tag = $32 - 10 - 5 = 17$ bits. Check $10 + 5 + 17 = 32 \checkmark$.
- Q4 20.** Block = $8\text{ words} \times 4\text{ B} = 32\text{ B} = 2^5 \Rightarrow$ offset 5 bits. Lines = $16\text{ KB}/32\text{ B} = 512 = 2^9$. Sets = $512/4 = 128 = 2^7 \Rightarrow$ index 7 bits. Address = $4\text{ GB} = 2^{32} \Rightarrow 32$ bits. Tag = $32 - 7 - 5 = 20$ bits.
- Q5 (D).** Blocks = $4\text{ K words} / 64\text{ words} = 64$ blocks. Sets = $64/4 = 16 = 2^4 \Rightarrow$ SET field 4 bits. WORD field = $\log_2 64 = 6$ bits. So SET, WORD = 4, 6.
- Q6 (C).** Two-way, four blocks $\Rightarrow$ 2 sets of 2 lines each; set = block mod 2. Trace: 8 (set 0, miss), 12 (set 0, miss), 0 (set 0, miss — evict LRU 8), 12 (hit), 8 (miss — evict LRU 0). Total = 4 misses.
- Q7 14.** Average read time = $0.8 \times 5 + 0.2 \times 50 = 4 + 10 = 14\text{ ns}$. (Here the miss time 50 ns is the total time on a miss, not a penalty added to the hit time.)
- Q8 1.68.** Instruction fetches: $90 \times 1 + 10 \times 5 = 140\text{ ns}$. Reads: $54 \times 1 + 6 \times 5 = 84\text{ ns}$. Writes: $36 \times 2 + 4 \times 10 = 112\text{ ns}$. Total = $140 + 84 + 112 = 336\text{ ns}$ over $100 + 60 + 40 = 200$ accesses $\Rightarrow 336/200 = 1.68\text{ ns}$.
- Q9 176.** Read-hit: $0.8 \times 0.9 \times 50 = 36$. Read-miss: $0.8 \times 0.1 \times 500 = 40$. Write (write-through updates both memories simultaneously, so every write costs the full main-memory time): $0.2 \times 500 = 100$. Total = $36 + 40 + 100 = 176\text{ ns}$.
- Q10 11.85.** Hierarchical: $0.95 \times 10 + 0.05 \times 0.85 \times 20 + 0.05 \times 0.15 \times 200 = 9.5 + 0.85 + 1.5 = 11.85\text{ ns}$. (Weight check: $0.95 + 0.0425 + 0.0075 = 1 \checkmark$.)

- Q11 9 and 10.** FIFO with 3 lines: misses at references 1, 2, 3, 4, 1, 2, 5, then hits at 1, 2, misses at 3, 4, hit at 5 $\Rightarrow$ 9 misses, 3 hits. FIFO with 4 lines: misses at 1, 2, 3, 4, hits at 1, 2, then misses at 5, 1, 2, 3, 4, 5 $\Rightarrow$ 10 misses, 2 hits. The larger cache misses *more* — this is Belady's anomaly.
- Q12 6.** OPT evicts the resident block whose *next* reference is furthest in the future. On the Week 9 trace, OPT scores 6 hits / 7 misses (hit ratio $6/13 = 46.2\%$), the ceiling no real policy can beat.
- Q13 5.** LFU evicts the line with the smallest reference count (ties broken by age). On the same string it scores 5 hits / 8 misses (38.5%), beating LRU (4 hits) and FIFO (3 hits) but below OPT's 6.